Kamal Sundar SK

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EXECUTIVE SUMMARY

Computer Science Engineer with experience building payment systems, agentic AI applications, and security-focused software, with a consistent focus on failure handling, audit integrity, and system correctness. Published research in IEEE and Taylor & Francis. Research Engineering Intern at IIT Madras CyStar.

PROFESSIONAL EXPERIENCE

Centre for Cybersecurity, Trust and Reliability (CyStar), IIT Madras Apr 2026 - Jun 2026
Research Intern

- Built production payment integration for a live SaaS platform using Razorpay, implementing idempotent webhook processing and atomic PostgreSQL transactions to guarantee zero duplicate entitlements and full state consistency under concurrent load.
- Designed and implemented CLGR (Compiled Locality Graph Retrieval), a graph-grounded document QA architecture achieving sub-20ms retrieval on 2,500+ entity graphs.

Software Testing (uTest), *Product tester* Oct 2025

- Executed a software testing cycle for a consumer web application, identifying 5+ usability issues and functional bugs.
- Delivered detailed bug reports with clear reproduction steps, screenshots, and logs, earning positive feedback for accuracy and clarity, and contributing to a faster triage process 15% improvement in resolution time

PUBLICATIONS

A Predictive IoT Framework for Continuous Vehicle Emission Monitoring Using Ensemble CatBoost and XGBoost Learning. May 2026
IEEE

- Proposed GreenPulse AI, a multi-factor IoT emission monitoring system using ensemble ML to forecast vehicle emissions one hour ahead with 93% accuracy.

Hybrid Neural Network Approach Using LSTM and XGBoost for Network Intrusion Detection and Classification Aug 2025
Taylor & Francis, Scopus Indexed

- Engineered a hybrid LSTM-XGBoost architecture for real-time network intrusion detection, reducing false positive rates by 30%

A Hidden Markov State Approach for Multistage Attack Detection in AI Agents for RCE and Prompt Injection Mar 2026
Under Review, Scopus-indexed journal

- HMM-based stateful security layer for AI agents detecting adversarial intent across multi-turn interactions with 92% precision against RCE and prompt injection.

A Four-Layer Semantic-Merkle Framework for Runtime Detection of Adversarial Tool Invocations in MCP-Based AI Agents
researched about multi stage tamper prevention in multi agent environements and preservation of model context protocoal (not published)

PROJECTS

SentinelStream — Real-Time Fraud Intelligence, *AlgoFest Hackathon*

- Architected and implemented a real-time fraud detection pipeline using Docker, Kafka, and Apache Flink, reducing fraudulent transaction identification time by 60%.
- Integrated a FastAPI-based model serving API to provide real-time fraud scores, leveraging Redis for feature retrieval and a pre-trained model for inference, improving detection accuracy by 25% and enabling sub-second response times.

Network intrusion detection using tensorflow

- Built a deep neural network with TensorFlow for network intrusion detection, achieving an 85% accuracy rate in identifying and classifying malicious activities.
- Optimized model performance through hyperparameter tuning, ensembling, and regularization, enhancing detection effectiveness by 83%.

NeuroCare-AI 🔗, *Nexathon-25-VIT*
- Developed a foundational AI model for diabetic retinopathy diagnosis, improving accuracy by 75% and efficiency by 89% through deep learning techniques.
- Enhanced chatbot functionality to support elderly mental health, improving accessibility, user experience, and engagement by 90%.

ClaimwiseAI 🔗, *Bajaj Finserv hackrx*
- Developed an LLM-based semantic search system, improving query accuracy by 35% and reducing response time by 40%
- Designed a dual-LLM architecture with embeddings for query matching and response generation, boosting relevant answer retrieval by 30%

Comtex (fake news detection), *nexus labs*
• Built modular backend architecture in Python and integrated ML models via REST APIs.
• Optimized data ingestion and preprocessing pipelines using **Kafka and FastAPI**, enabling real-time classification.

Claimwise - AI, *Bajaj Finserv hackrx*
• Developed a full-stack Python-based semantic search engine with efficient query indexing and retrieval logic.
• Implemented containerized deployment with **Docker** and maintained CI/CD for model updates.

project visior 🔗, *interactgen hackfest* Dec 2025
- Built end-to-end autonomous desktop assistant with Electron + TypeScript overlay, FastAPI backend, and React + Vite frontend, integrating real-time vision processing and multimodal speech interaction for intelligent screen navigation
- Engineered full-stack orchestration using Python AsyncIO cdesktop automation, achieving 85% feature completion with persistent context memory

Investigative Journalism Intelligence Engine 🔗, *ET HACKATHON 2026*
- Built an event-driven agentic pipeline for real-time news ingestion with deterministic guardrails and parallel dispatch across multi-stage autonomous investigations.

Agentic Personal Data Exposure Detection System 🔗
- Designed a 7-phase autonomous detection pipeline with phase-gated execution and prompt injection sanitization, validated against 8 simulated attack scenarios including break-glass authorization bypass attempts.

Autonomous Research Agent with trust system 🔗, *github hackathon 2026*
- Developed an autonomous research agent with tamper-evident event sourcing using SHA-256 verification across tool interactions, ensuring integrity of execution flows
- Engineered a two-tier memory system combining context windowing and embedding-based retrieval, with automated compression triggered after 20 exchanges to optimize performance

EDUCATION

Shiv Nadar University , Chennai Jul 2026 | chennai , India
M.Tech - Computer Science And Engineering

SRM Institute of Science and Technology, Vadapalani 2022 - 2026 | Chennai, India
B.Tech - Computer Science and Business Systems
GPA: 9.20

St. John's English School & Junior College, Besant Nagar Chennai, India
High School Diploma (Class 12)
Score:90.2%

SKILLS

languages — Python, Java, C, C++,typescript,Go

AI/ML & Security: — PyTorch, TensorFlow, Scikit-learn, LangGraph, GraphRAG

Infrastructure — Docker, Apache Kafka, Redis, FastAPI, Node.js

Database: — PostgreSQL, MongoDB, MySQL, Neo4j, SQLite, pgvector

Soft Skills — Problem-Solving | Communication | Adaptability | Leadership

Concepts: — SDLC, CI/CD, System Optimization, Debugging, Open-Source Contribution

CERTIFICATES

Generative Al Virtual Internship
Focused on Generative AI and related concepts.

Google generative AI Vitrual Intership-Eduskills
Virtual internship on Advanced AI concepts and applications.